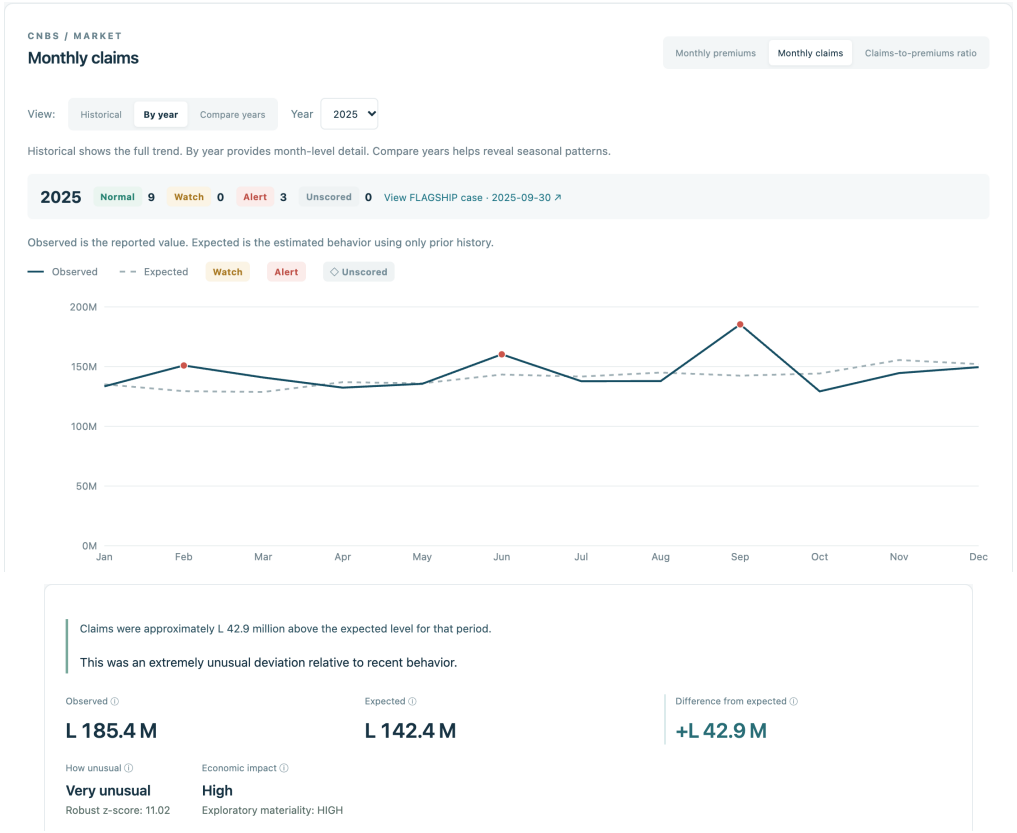
ACTUARIAL SIGNAL RADAR

Evolus Business Agents on AMD

We detect first.
AI investigates second.
Humans decide.

Diego Turcios

REAL PRODUCT / CNBS HONDURAS



Dashboards tell us what happened. We ask whether it was unusual.

TRADITIONAL

“What happened?”

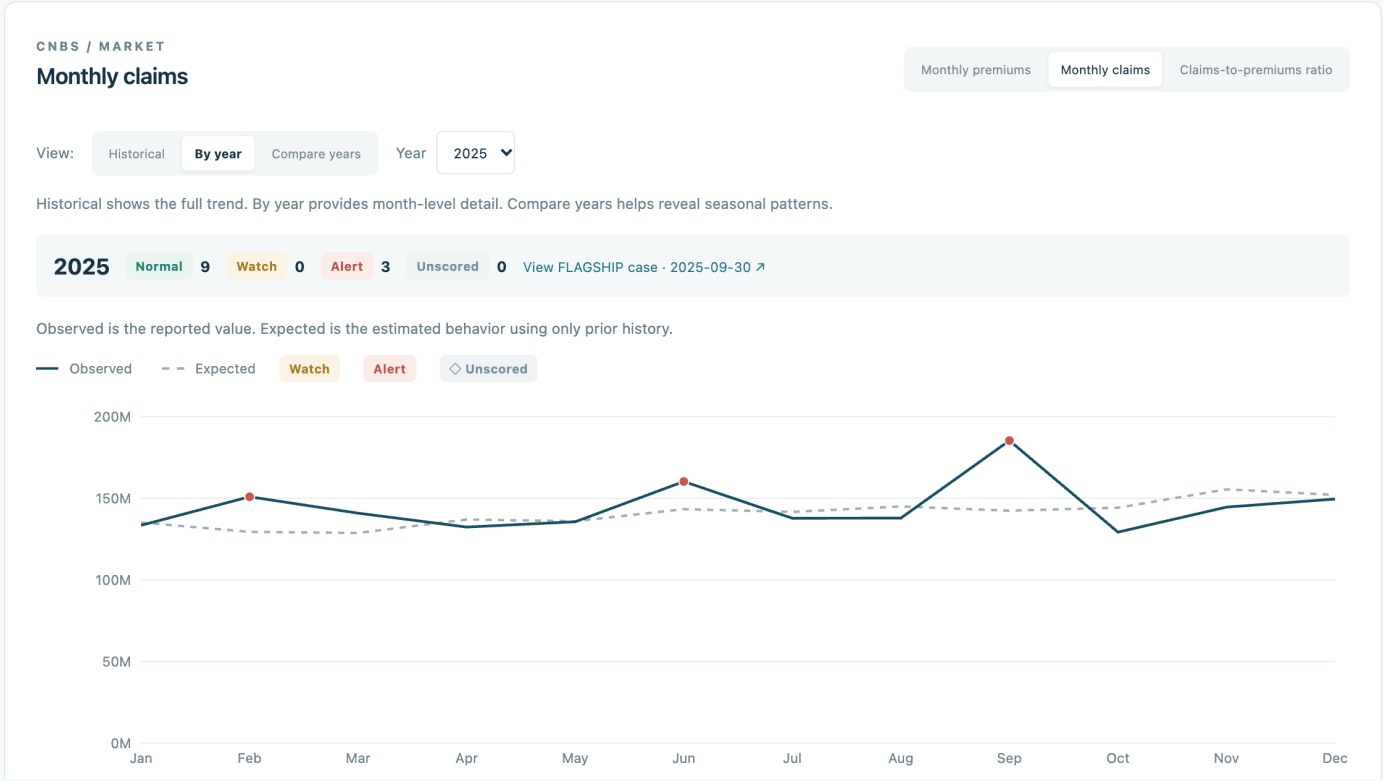
A better next question

What was expected?

How different was reality?

Was that difference unusual?

What should we investigate?



MARCH 2025 · CLAIMS-TO-PREMIUMS RATIO

Raw NORMAL → adjusted ALERT

63.91% vs 53.05% · +10.86 pp · z 8.61

Detection happens before AI

We learn what normally happens. Then we measure whether reality was unusually different.

01 — Prior history
Previously available monthly observations

→

02 — Trend + seasonality
Direction + recurring monthly patterns

→

03 — Expected
A reference for this reporting month

→

04 — Difference
Observed − Expected (the residual)

→

05 — Robust Z
How unusual is that difference?

NORMAL | WATCH | ALERT

Primary: 12-month robust z-score on residuals · Complementary: standard Z, EWMA, two-sided CUSUM

NOV 2020 Large spike, but UNSCORED: insufficient residual history. No marker ≠ NORMAL.

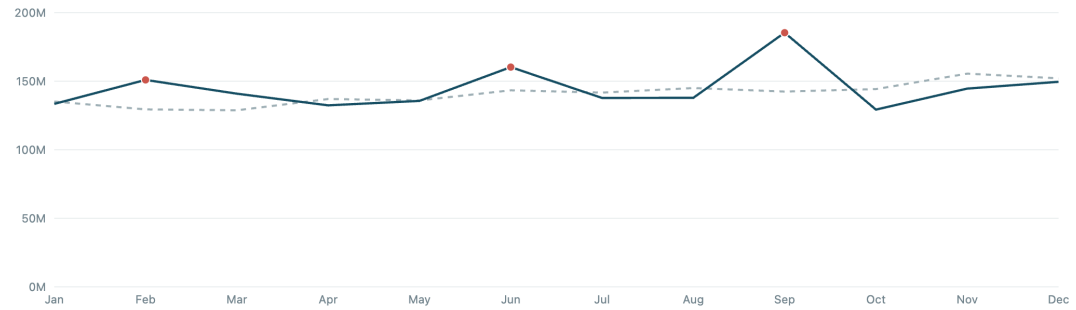
September 2025 triggered the signal

L185.4M
OBSERVED MONETARY CLAIMS

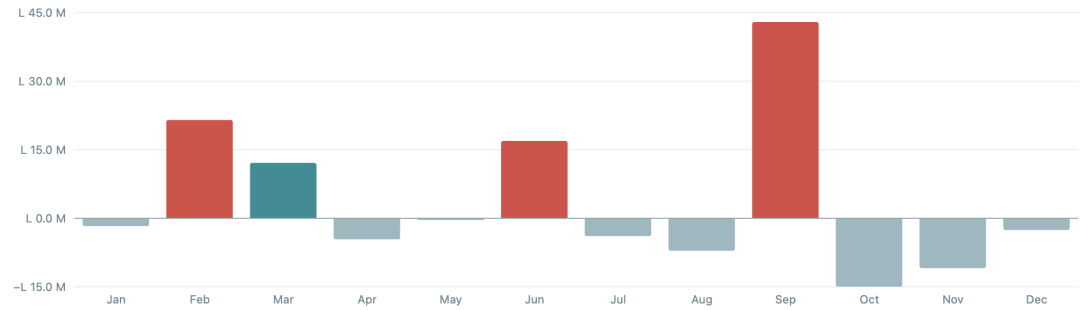
L142.4M
EXPECTED

+L42.9M
ABOVE EXPECTED

OBSERVED VS EXPECTED · 2025



DIFFERENCE FROM EXPECTED · LEMPIRAS



ALERT **+30.15%** Robust Z 11.02 · HIGH exploratory materiality

Reported: 2025-09-30 → Available for analysis (simulated): 2025-12-01

Detection and investigation are different jobs

STATISTICS

Is something unusual?

→

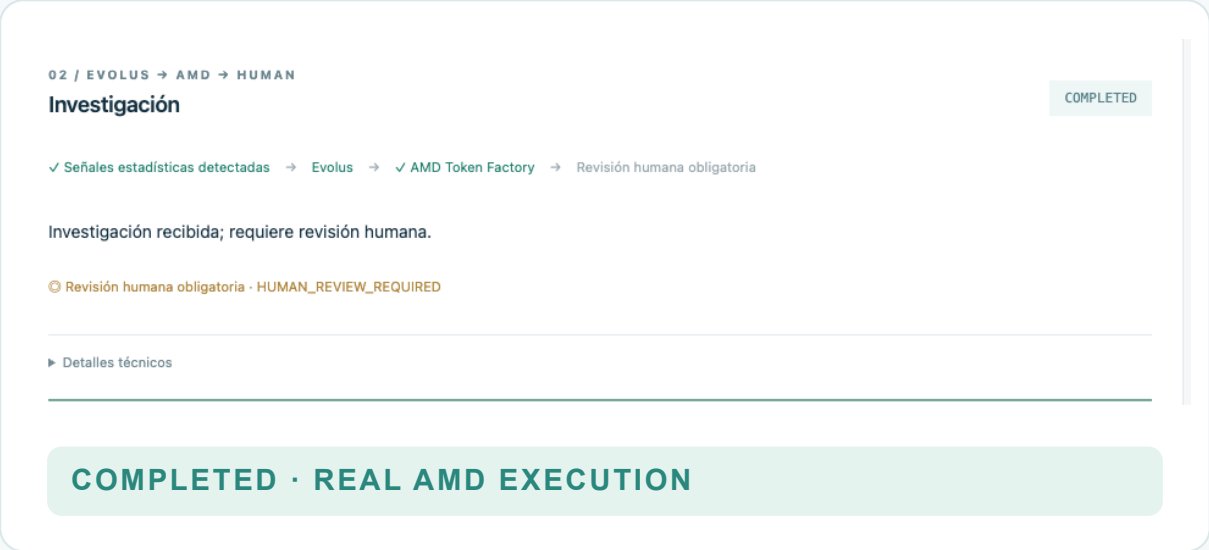
AI

What should we investigate?

→

HUMAN

What does it mean? What should we do?



Recorded real public run · Spanish output · Human review preserved

The AI never creates or overrides the alert.

Summarize validated evidence

Propose testable hypotheses

Identify missing information

Recommend the next checks

HUMAN_REVIEW_REQUIRED

A business-agent workflow, not just a chatbot

01 / HANDOFF

Validated signal

Immutable values
Existing ALERT
Quality + sensitivity context

→

EVOLUS

Actuarial Signal Investigation

Workflow orchestration
Knowledge Library context
Structured-response validation
Mandatory human review

6 indexed documents · checksums verified

→

02 / INFERENCE

AMD Token Factory

DeepSeek-V4-Flash

OpenAI-compatible
Chat Completions API

Structured investigation → HUMAN_REVIEW_REQUIRED

Broad market coverage. Limited temporal granularity.

HONDURAS

Available motor-insurance market

2018–2026

Historical source snapshot

≈100 months

Per primary time series

An interpretable statistical baseline

Complex detectors could overfit a small temporal sample.

AUDITABLE

REPRODUCIBLE

TRANSPARENT

NEXT, IF RICHER DATA BECOMES AVAILABLE

Claim-level frequency + severity + richer variables → benchmark ML against this baseline.

From public data to a production investigation



VERIFIED IN RECORDED RUNS

- ✓ Public dashboard
- ✓ Railway production API
- ✓ Evolus workflow
- ✓ AMD Token Factory
- ✓ DeepSeek-V4-Flash
- ✓ Mandatory human review

PARTIAL / VERIFIED FOR CLAIMS CASE

Claims completed in ~154 seconds. Ratio retry: “Missing model execution envelope”. Safeguards were not weakened.

We detect first. AI investigates second. Humans decide.

LIVE DEMO actuarial-signal-radar.diegoturciostc.workers.dev

GITHUB github.com/DiegoTc/o-Actuarial-Signal-Radar